Hidden Markov Models

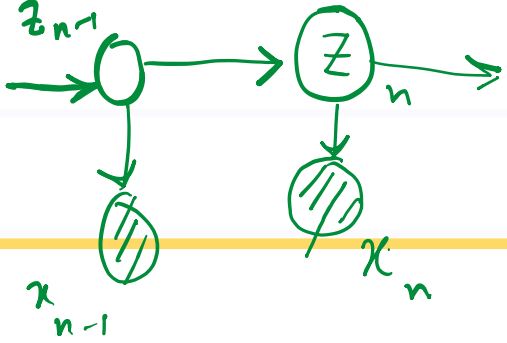
EE698V - Machine Learning for Signal Processing

Vipul Arora

Hidden Markov Models

Reference: PRML Section 13.2

Joint Distribution



$$p(x_N, x_{N-1}, \dots, x_1, z_N, z_{N-1}, \dots, z_1) = \underbrace{p(z_1)}_{\text{initial state}} \underbrace{\prod_{n=2}^{N} p(z_n | z_{n-1})}_{\text{transition probability}} \underbrace{\prod_{n=1}^{N} p(x_n | z_n)}_{\text{emission probability}}$$

Example:

- $x_n \in \mathbb{R}$ is the temperature on day n
- $z_n \in \{1,2,\dots,6\}$ is one of the 6 seasons

Estimating the parameters

Reference: PRML Section 13.2

ML estimation

$$\theta^* = \underset{\theta}{\operatorname{argmax}}\ \mathcal{L}(\theta), \qquad \text{where } \mathcal{L}(\theta) = \log p(x_1, \dots, x_N \mid \theta)$$

$$\mathcal{L}(\theta) = \log\left\{ \sum_{z_1,\dots,z_N} p(x_1,\dots,x_N \mid z_1,\dots,z_N,\theta) \cdot \frac{p(z_1,\dots,z_N \mid \theta)}{q(z_1,\dots,z_N \mid x_1,\dots,x_N,\theta^{old})} \times q(z_1,\dots,z_N \mid x_1,\dots,x_N,\theta^{old}) \right\}$$

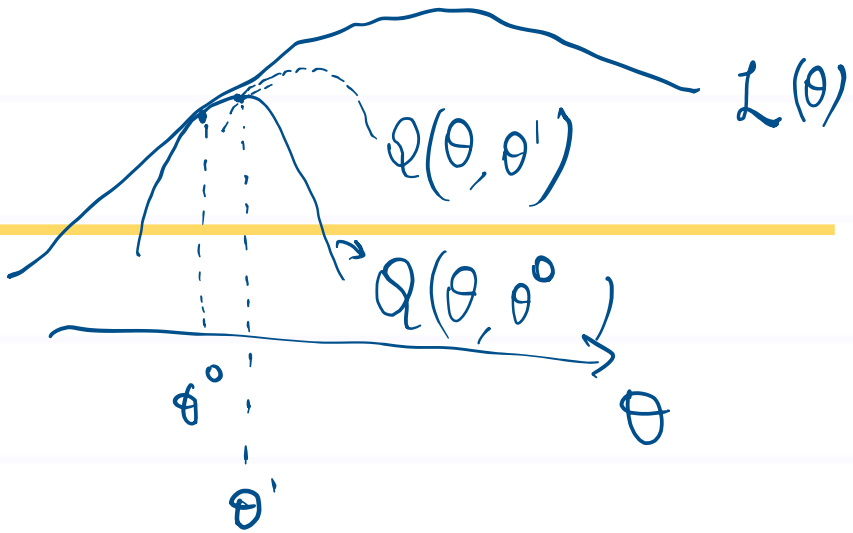
$$\geq \sum_{z_1,\dots,z_N} q(z_1,\dots,z_N \mid x_1,\dots,x_N,\theta^{old}) \log\left\{ p(x_1,\dots,x_N \mid z_1,\dots,z_N,\theta)\, p(z_1,\dots,z_N \mid \theta) \right\}$$

$$= Q(\theta, \theta^{old})$$

Take $q(z_1,\dots,z_N \mid x_1,\dots,x_N,\theta^{old}) = \prod_{n=1}^{N} q(z_n \mid z_{n-1}, x_1,\dots,x_N,\theta^{old})$

1st order Markov assumption

EM Algorithm



- E step
  - $q(z_1, \dots, z_N | x_1, \dots, x_N, \theta) = p\left(z_1, \dots, z_N \middle| x_1, \dots, x_N, \theta^{\text{old}}\right)$
- M step
  - $\theta \leftarrow \text{argmax}_\theta \mathbb{E}[\log\left(p(x_1, \dots, x_N | z_1, \dots, z_N, \theta) p(z_1, \dots, z_N | \theta)\right)]$
- M step is easy. But E is a bit involved - it uses Forward Backward algorithm or Baum Welch algorithm
- Could you suggest a simpler approach with some gross approximation? Hint: you know Viterbi algorithm

Recap

$$z_1, \dots, z_N = \underset{z_1, \dots, z_N}{\operatorname{argmax}} \prod_{n=1}^{N} p(x_n | z_n) \prod_{n=2}^{N} p(z_n | z_{n-1}) \; \pi(z_1)$$

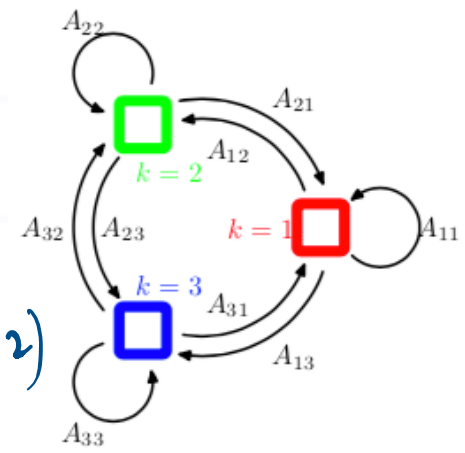
$$= \underset{z_1, \dots, z_N}{\operatorname{argmax}} \left\{ \sum_{n=1}^{N} \log p(x_n | z_n) + \sum_{n=2}^{N} \log p(z_n | z_{n-1}) + \log \pi(z_1) \right\}$$

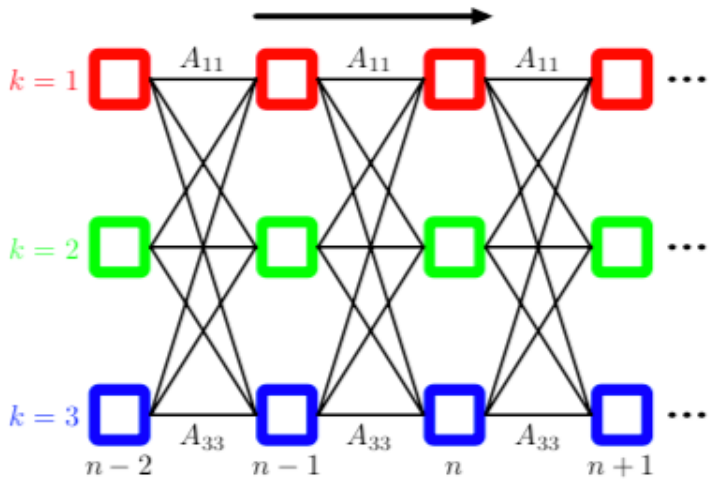
$p(x_n | z_n = 1) \; \gamma(z_n = 1)$

$p(x_n | z_n = 2) \; \gamma(z_n = 2)$

$p(x_n | z_n = 3) \; \gamma(z_n = 3)$

$\vdots$





$x_{n-2} \quad x_{n-1} \quad x_n$

EM Algorithm: with single best path

- E step
  - $q(z_1, \dots, z_N | x_1, \dots, x_N, \theta) =$

    $\max_{z_1,\dots,z_N} p\left(z_1, \dots, z_N \middle| x_1, \dots, x_N, \theta^{\text{old}}\right)$
  - $z_1^*, \dots, z_N^* = \text{argmax}_{z_1,\dots,z_N} p\left(z_1, \dots, z_N \middle| x_1, \dots, x_N, \theta^{\text{old}}\right)$
- M step
  - $\theta \leftarrow \text{argmax}_\theta \log\left(p(x_1, \dots, x_N | z_1^*, \dots, z_N^*, \theta) p(z_1^*, \dots, z_N^* | \theta)\right)$

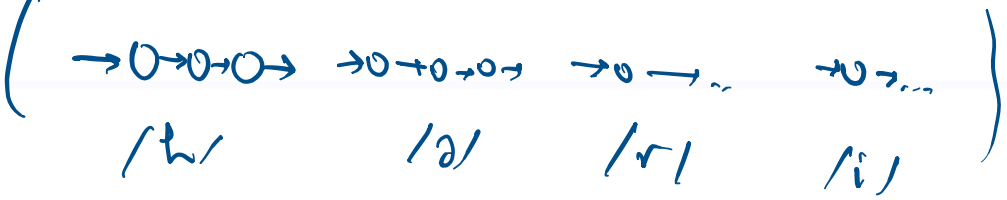
EM Algorithm: with single best path

- Given θ^{old}, estimate the best path $z_1, \ldots, z_N$
- Given the best path, estimate θ
- Iterate until convergence

- Saves a lot of computations (and calculations too!)
- Performs well with large data
- Popularly used for ASR training

/h/ + /ə/ + /r/ + /i/

Summary



hari

You learnt:

- Making models for sequential/temporal data
- Finding time alignment of states (Viterbi), given the model, the observation and the corresponding sequence of states: Forced Alignment
  - What if the sequence of states is not given? Decoding
- Estimating the parameters of HMM, given the observations and the corresponding sequence of states
  - What if only states, and not their sequence, are given? e.g. Unsupervised ASR